

Graph Library: Graph Containers

Document #: **P3131r1**
Date: 2024-02-05
Project: Programming Language C++
Audience: Library Evolution
SG19 Machine Learning
SG14 Game, Embedded, Low Latency
SG6 Numerics
Revises: P3131r0

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1 Getting Started

This paper is one of several interrelated papers for a proposed Graph Library for the Standard C++ Library. The Table 1 describes all the related papers.

Paper	Status	Description
P1709	Inactive	Original proposal, now separated into the following papers.
P3126	Active	Overview , describes the big picture of what we are proposing.
P3127	Active	Background and Terminology provides the motivation, theoretical background, and terminology used across the other documents.
P3128	Active	Algorithms covers the initial algorithms as well as the ones we'd like to see in the future.
P3129	Active	Views has helpful views for traversing a graph.
P3130	Active	Graph Container Interface is the core interface used for uniformly accessing graph data structures by views and algorithms. It is also designed to easily adapt to existing graph data structures.
P3131	Active	Graph Containers describes a proposed high-performance <code>compressed_graph</code> container. It also discusses how to use containers in the standard library to define a graph, and how to adapt existing graph data structures.

Table 1: Graph Library Papers

Reading them in order will give the best overall picture. If you're limited on time, you can use the following guide to focus on the papers that are most relevant to your needs.

Reading Guide

- If you're **new to the Graph Library**, we recommend starting with the *Overview* paper (P3126) to understand the focus and scope of our proposals.
- If you want to **understand the theoretical background** that underpins what we're doing, you should read the *Background and Terminology* paper (P3127).
- If you want to **use the algorithms**, you should read the *Algorithms* paper (P3128) and *Graph Containers* paper (P3131).
- If you want to **write new algorithms**, you should read the *Views* paper (P3129), *Graph Container Interface* paper (P3130), and *Graph Containers* paper (P3131). You'll also want to review existing implementations in the reference library for examples of how to write the algorithms.
- If you want to **use your own graph container**, you should read the *Graph Container Interface* paper (P3130) and *Graph Containers* paper (P3131).

2 Revision History

P3131r0

- Split from P1709r5. Added *Getting Started* section.
- Move text for graph data structures created from std containers from Graph Container Interface to Container Implementation paper.
- GCI overloads are no longer required for adjacency lists constructed with standard containers. Data structures that follow the pattern `random_access_range<forward_range<integral>>` and `random_access_range<forward_range<tuple<integral, ...>>>` are automatically recognized as an adjacency list, including containers from non-standard libraries. The `integral` value is used as the `target_id`.

P3131r1

- Added feature summary of `compressed_graph` beyond the typical CSR implementation.
- Added complexity for `num_edges(g)` and `has_edge(g)` functions in `compressed_graph`.
- Add constructors to `compressed_graph` to complement the removal of the load functions from [P3130r1 Graph Container Interface](#). An optional `partition_start_ids` parameter is also included.

3 Naming Conventions

Table 2 shows the naming conventions used throughout the Graph Library documents.

Template Parameter	Type Alias	Variable Names	Description
G			Graph
	<code>graph_reference_t<G></code>	<code>g</code>	Graph reference
GV		<code>val</code>	Graph Value, value or reference
V	<code>vertex_t<G></code>		Vertex
	<code>vertex_reference_t<G></code>	<code>u,v,x,y</code>	Vertex reference. <code>u</code> is the source (or only) vertex. <code>v</code> is the target vertex.
VId	<code>vertex_id_t<G></code>	<code>uid,vid,seed</code>	Vertex id. <code>uid</code> is the source (or only) vertex id. <code>vid</code> is the target vertex id.
VV	<code>vertex_value_t<G></code>	<code>val</code>	Vertex Value, value or reference. This can be either the user-defined value on a vertex, or a value returned by a function object (e.g. <code>VVF</code>) that is related to the vertex.
VR	<code>vertex_range_t<G></code>	<code>ur,vr</code>	Vertex Range
VI	<code>vertex_iterator_t<G></code>	<code>ui,vi</code>	Vertex Iterator. <code>ui</code> is the source (or only) vertex.
		<code>first,last</code>	<code>vi</code> is the target vertex.
VVF		<code>vvf</code>	Vertex Value Function: <code>vvf(u) → vertex value</code> , or <code>vvf(uid) → vertex value</code> , depending on requirements of the consume algorithm or view.
VProj		<code>vproj</code>	Vertex descriptor projection function: <code>vproj(x) → vertex_descriptor<VId,VV></code> .
	<code>partition_id_t<G></code>	<code>pid</code>	Partition id.
		<code>P</code>	Number of partitions.
PVR	<code>partition_vertex_range_t<G></code>	<code>pur,pvr</code>	Partition vertex range.
E	<code>edge_t<G></code>		Edge
	<code>edge_reference_t<G></code>	<code>uv,vw</code>	Edge reference. <code>uv</code> is an edge from vertices <code>u</code> to <code>v</code> . <code>vw</code> is an edge from vertices <code>v</code> to <code>w</code> .
EId	<code>edge_id_t<G></code>	<code>eid,uvid</code>	Edge id, a pair of vertex_ids.
EV	<code>edge_value_t<G></code>	<code>val</code>	Edge Value, value or reference. This can be either the user-defined value on an edge, or a value returned by a function object (e.g. <code>EVF</code>) that is related to the edge.
ER	<code>vertex_edge_range_t<G></code>		Edge Range for edges of a vertex
EI	<code>vertex_edge_iterator_t<G></code>	<code>uvi,vwi</code>	Edge Iterator for an edge of a vertex. <code>uvi</code> is an iterator for an edge from vertices <code>u</code> to <code>v</code> . <code>vwi</code> is an iterator for an edge from vertices <code>v</code> to <code>w</code> .
EVF		<code>evf</code>	Edge Value Function: <code>evf(uv) → edge value</code> , or <code>evf(eid) → edge value</code> , depending on the requirements of the consuming algorithm or view.
EProj		<code>eproj</code>	Edge descriptor projection function: <code>eproj(x) → edge_descriptor<VId,Sourced,EV></code> .

Table 2: Naming Conventions for Types and Variables

4 compressed_graph Graph Container

`compressed_graph` is a graph container being proposed for the standard library. It is a high-performance data structure that uses [Compressed Sparse Row](#) (CSR) format to store its vertices, edges and associated values. Once constructed, vertices and edges cannot be added or deleted but values on vertices and edges can be modified.

There are a number of features added beyond the typical CSR implementation:

- **User-defined values** The typical CSR implementation stores values on edges (columns) by defining the `EV` template parameter. `compressed_graph` extends that to also allow values on vertices (rows) and the graph itself by defining the `VV` and `GV` template arguments respectively. If a type is void, no memory overhead is incurred.
- **Index type sizes** The size of the integral indexes into the internal vertex (row) and edge (column) structures can be controlled by the `VId` and `EIndex` template arguments respectively to give a balance between capacity, memory usage and performance.
- **Multi-partite graphs** The vertices can optionally be partitioned into multiple partitions by passing the starting vertex id of each partition in the `partition_start_ids` argument in the constructors. If no partitions are specified, the graph is single-partite.

The listings in the following sections show the prototypes for the `compressed_graph` when the graph value type `GV` is non-`void` (section 4.1) and a class template specialization when it is `void` (section 4.2).

Only the constructors and destructor shown for `compressed_graph` are public. All other types and functions related to the graph are only accessible through the types and functions in the Graph Container Interface.

vertex_id assignment: Contiguous	<code>has_edge(g)</code> $O(1)$	Append vertices? No
Vertices range: Contiguous	<code>num_edges(g)</code> $O(1)$	Append edges? No
Edge range: Contiguous	<code>partition_id(g,uid)</code> $O(\log(P+1))$	Partions? Yes

P is the number of partitions and is expected to be small, e.g. $P = 2$ for bipartite and $P \leq 10$ for typical multi-partite graphs.

[PHIL: Add `operator[] (vertex_id_t<G>) ?`]

4.1 compressed_graph when `GV` is not `void`

```
template <class EV,
          class VV,
          class GV,
          integral VId=uint32_t,
          integral EIndex=uint32_t,
          class Alloc=allocator<VId>>
class compressed_graph {
public: // Construction/Destruction/Assignment
    constexpr compressed_graph() = default;
    constexpr compressed_graph(const compressed_graph&) = default;
    constexpr compressed_graph(compressed_graph&&) = default;
    constexpr ~compressed_graph() = default;

    constexpr compressed_graph& operator=(const compressed_graph&) = default;
    constexpr compressed_graph& operator=(compressed_graph&&) = default;

    // compressed_graph( alloc)
    // compressed_graph(gv&, alloc)
    // compressed_graph(gv&&, alloc)

    constexpr compressed_graph(const Alloc& alloc);
    constexpr compressed_graph(const graph_value_type& value, const Alloc& alloc = Alloc());
```

```

constexpr compressed_graph(graph_value_type&& value, const Alloc& alloc = Alloc());

// compressed_graph(erng, eprojection, alloc)
// compressed_graph(gv&&, erng, eprojection, alloc)
// compressed_graph(gv&&, erng, eprojection, alloc)

template <ranges::forward_range ERng, ranges::forward_range PartRng, class EProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, Vid, EV> &&
         convertible_to<ranges::range_value_t<PartRng>, Vid>
constexpr compressed_graph(const ERng& erng,
                           EProj eprojection,
                           const PartRng& partition_start_ids = vector<Vid>(),
                           const Alloc& alloc = Alloc());

template <ranges::forward_range ERng, ranges::forward_range PartRng, class EProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, Vid, EV> &&
         convertible_to<ranges::range_value_t<PartRng>, Vid>
constexpr compressed_graph(const graph_value_type& value,
                           const ERng& erng,
                           EProj eprojection,
                           const PartRng& partition_start_ids = vector<Vid>(),
                           const Alloc& alloc = Alloc());

template <ranges::forward_range ERng, ranges::forward_range PartRng, class EProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, Vid, EV> &&
         convertible_to<ranges::range_value_t<PartRng>, Vid>
constexpr compressed_graph(graph_value_type&& value,
                           const ERng& erng,
                           EProj eprojection,
                           const PartRng& partition_start_ids = vector<Vid>(),
                           const Alloc& alloc = Alloc());

// compressed_graph(erng, vrng, eprojection, vprojection, alloc)
// compressed_graph(gv&&, erng, vrng, eprojection, vprojection, alloc)
// compressed_graph(gv&&, erng, vrng, eprojection, vprojection, alloc)

template <ranges::forward_range ERng,
         ranges::forward_range VRng,
         ranges::forward_range PartRng,
         class EProj = identity,
         class VProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, Vid, EV> &&
         copyable_vertex<invoke_result<VProj, ranges::range_value_t<VRng>>, Vid, VV> &&
         convertible_to<ranges::range_value_t<PartRng>, Vid>
constexpr compressed_graph(const ERng& erng,
                           const VRng& vrng,
                           EProj eprojection = {},
                           VProj vprojection = {},
                           const PartRng& partition_start_ids = vector<Vid>(),
                           const Alloc& alloc = Alloc());

template <ranges::forward_range ERng,
         ranges::forward_range VRng,
         ranges::forward_range PartRng,
         class EProj = identity,
         class VProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, Vid, EV> &&
         copyable_vertex<invoke_result<VProj, ranges::range_value_t<VRng>>, Vid, VV> &&
         convertible_to<ranges::range_value_t<PartRng>, Vid>

```

```

constexpr compressed_graph(const graph_value_type& value,
                           const ERng& erng,
                           const VRng& vrng,
                           EProj eprojection = {},
                           VProj vprojection = {},
                           const PartRng& partition_start_ids = vector<VId>(),
                           const Alloc& alloc = Alloc());

template <ranges::forward_range ERng,
          ranges::forward_range VRng,
          ranges::forward_range PartRng,
          class EProj = identity,
          class VProj = identity>
requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, VId, EV> &&
         copyable_vertex<invoke_result<VProj, ranges::range_value_t<VRng>>, VId, VV> &&
         convertible_to<ranges::range_value_t<PartRng>, VId>
constexpr compressed_graph(graph_value_type&& value,
                           const ERng& erng,
                           const VRng& vrng,
                           EProj eprojection = {},
                           VProj vprojection = {},
                           const PartRng& partition_start_ids = vector<VId>(),
                           const Alloc& alloc = Alloc());

constexpr compressed_graph(const initializer_list<copyable_edge_t<VId, EV>>& ilist,
                           const Alloc& alloc = Alloc());
};

```

4.2 compressed_graph specialization when GV is void

When GV is void the number of constructors decreases significantly as shown in the following listing.

```

template <class EV,
          class VV,
          integral VId=uint32_t,
          integral EIndex=uint32_t,
          class Alloc=allocator<VId>>
template <class EV, class VV, integral VId, integral EIndex, class Alloc>
class compressed_graph<EV, VV, void, VId, EIndex, Alloc>
public: // Construction/Destruction
    constexpr compressed_graph() = default;
    constexpr compressed_graph(const compressed_graph&) = default;
    constexpr compressed_graph(compressed_graph&&) = default;
    constexpr ~compressed_graph() = default;

    constexpr compressed_graph& operator=(const compressed_graph&) = default;
    constexpr compressed_graph& operator=(compressed_graph&&) = default;

    // edge-only construction
    template <ranges::forward_range ERng, class EProj = identity>
    requires copyable_edge<invoke_result<EProj, ranges::range_value_t<ERng>>, VId, EV>
    constexpr compressed_graph(const ERng& erng,
                              EProj eprojection = identity(),
                              const Alloc& alloc = Alloc());

    // edge and vertex value construction
    template <ranges::forward_range ERng,
              ranges::forward_range VRng,
              ranges::forward_range PartRng,

```

```

        class EProj = identity,
        class VProj = identity>
constexpr compressed_graph(const ERng& erng,
                           const VRng& vrng,
                           EProj eprojection = {},
                           VProj vprojection = {},
                           const PartRng& partition_start_ids = vector<VId>(),
                           const Alloc& alloc = Alloc());

// initializer list using edge_descriptor<VId,true,void,EV>
constexpr compressed_graph(const initializer_list<copyable_edge_t<VId, EV>>& ilist,
                           const Alloc& alloc = Alloc());
};

```

4.3 compressed_graph description

[PHIL: Is it possible to support movable EV and VV types?]

1 *Mandates:*

- (1.1) — The `EV` template argument for an edge value must be a copyable type or `void`.
- (1.2) — The `VV` template argument for a vertex value must be a copyable type or `void`.
- (1.3) — When the `GV` template argument for a graph value is not `void` it can be movable or copyable. It must have a default constructor if it is not passed in a `compressed_graph` constructor.
- (1.4) — The `EProj` template argument must be a projection that returns a value of `copyable_edge<VId, true, EV>` type given a value of `erng`. If the value type of `ERng` is already a `copyable_edge<VId, true, EV>` type, then `EProj` can be `identity`.
- (1.5) — The `VProj` template argument must be a projection that returns a value of `copyable_vertex<VId, VV>` type, given a value of `vrng`. If the value type of `VRng` is already a `copyable_vertex<VId, VV>` type, then `VProj` can be `identity`.

2 *Preconditions:*

- (2.1) — The `VId` template argument must be able to store a value of $|V|+1$, where $|V|$ is the number of vertices in the graph. The size of this type impacts the size of the *edges*.
- (2.2) — The `EIndex` template argument must be able to store a value of $|E|+1$, where $|E|$ is the number of edges in the graph. The size of this type impact the size of the *vertices*.
- (2.3) — The `EProj` and `VProj` template arguments must be valid projections.
- (2.4) — The `partition_start_ids` range includes the starting vertex id for each partition. If it is empty, then the graph is single-partite and the number of partitions is 1. If it is not empty, then the number of partitions is the size of the range, where the first element must be 0 and all elements are in ascending order. A vertex id in the range must not exceed the number of vertices in the graph. Any violation of these conditions results in undefined behavior.

[PHIL: If duplicate `partition_start_ids` exist they create an empty partition with no vertices.]

3 *Effects:*

- (3.1) — When `EV`, `VV`, or `GV` are `void`, no extra memory overhead is incurred for that type.

4 *Remarks:*

- (4.1) — The `VId` and `EIndex` template arguments impact the capacity, internal storage requirements and performance. The default of `uint32_t` is sufficient for most graphs and provides a good balance between storage and performance.

The memory requirements are roughly,

$$|V| \times (\text{sizeof}(EIndex) + \text{sizeof}(VV)) + |E| \times (\text{sizeof}(VId) + \text{sizeof}(EV)) + \text{sizeof}(GV)$$

where $|V|$ is the number of vertices and $|E|$ is the number of edges in the graph. `sizeof void` is 0 when considering `sizeof` for `VV`, `EV`, and `GV`. Alignment and overhead for internal vectors are not included in this calculation.

(4.2) — The allocator passed to constructors is rebound for different types used by different internal containers.

5 Using Existing Graph Data Structures

Reasonable defaults have been defined for the GCI functions to minimize the amount of work needed to adapt an existing graph data structure to be used by the views and algorithms.

There are two cases supported. The first is for the use of standard containers to define the graph and the other is for a broader set of more complicated implementations.

5.1 Using Standard Containers for the Graph Data Structure

For example this we'll use `G = vector<forward_list<tuple<int,double>>>` to define the graph, where `g` is an instance of `G`. `tuple<int,double>` defines the `target_id` and `weight` property respectively. We can write loops to go through the vertices, and edges within each vertex, as follows.

```
using G = vector<forward_list<tuple<int,double>>>;
auto weight = [&g](edge_t& uv) { return get<1>(uv); }

G g;
load_graph(g, ...); // load some data

// Using GCI functions
for(auto&& [uid, u] : vertices(g)) {
    for(auto&& [vid, uv]: edges(g,u)) {
        auto w = weight(uv);
        // do something...
    }
}
```

Note that no function override was required and is a special case when the outer range is a `random_access_range` and inner inner range is a `forward_range`, and the value type of the inner range is either `integral` or `tuple<integral, ...>`. This extends to any range type. For instance, `boost::containers` can be used just as easily as `std` containers.

Function or Value	Concrete Type
<code>vertices(g)</code>	<code>vector<forward_list<tuple<int,double>>></code> (when <code>random_access_range<G></code>)
<code>u</code>	<code>forward_list<tuple<int,double>></code>
<code>edges(g,u)</code>	<code>forward_list<tuple<int,double>></code> (when <code>random_access_range<vertex_range_t<G>></code>)
<code>uv</code>	<code>tuple<int,double></code>
<code>edge_value(g,uv)</code>	<code>tuple<int,double></code> (when <code>random_access_range<vertex_range_t<G>></code>)
<code>target_id(g,uv)</code>	<code>integral</code> , when <code>uv</code> is either <code>integral</code> or <code>tuple<integral,...></code>

Table 3: Types When Using Standard Containers

5.2 Using Other Graph Data Structures

For other graph data structures more function overrides are required. Table 4 shows the common function overrides anticipated for most cases, keeping in mind that all functions can be overridden. When they are defined they must be in the same namespace as the data structures.

Function	Comment
<code>vertices(g)</code>	
<code>edges(g,u)</code>	
<code>target_id(g,uv)</code>	
<code>edge_value(g,uv)</code>	If edges have value(s) in the graph
<code>vertex_value(g,u)</code>	If vertices have value(s) in the graph
<code>graph_value(g)</code>	If the graph has value(s)

	When edges have the optional <code>source_id</code> on an edge
<code>source_id(g,uv)</code>	

	When the graph supports multiple partitions
<code>num_partitions(g)</code>	
<code>partition_id(g,u)</code>	
<code>vertices(g,u,pid)</code>	

Table 4: Common CPO Function Overrides

Acknowledgements

Phil Ratzloff's time was made possible by SAS Institute.

Portions of *Andrew Lumsdaine's* time was supported by NSF Award OAC-1716828 and by the Segmented Global Address Space (SGAS) LDRD under the Data Model Convergence (DMC) initiative at the U.S. Department of Energy's Pacific Northwest National Laboratory (PNNL). PNNL is operated by Battelle Memorial Institute under Contract DE-AC06-76RL01830.

Michael Wong's work is made possible by Codeplay Software Ltd., ISO CPP Foundation, Khronos and the Standards Council of Canada.

Muhammad Osama's time was made possible by Advanced Micro Devices, Inc.

The authors thank the members of SG19 and SG14 study groups for their invaluable input.